Docking EKF: equation set

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1 State, inputs and timing

$$\boldsymbol{x} = \begin{bmatrix} p_x \\ p_y \\ \theta \\ v_x \\ v_y \end{bmatrix} \in \mathbb{R}^5, \qquad \boldsymbol{u} = \begin{bmatrix} a_x \\ a_y \\ r \end{bmatrix} \in \mathbb{R}^3, \qquad T = \tfrac{1}{200}\ \text{s}. \tag{1}$$

$\boldsymbol{p}$ and $\boldsymbol{v}$ are the position and velocity of the chassis centre expressed in the target frame T; θ is the heading of the body x-axis relative to the target x-axis. a_x, a_y are the gravity-compensated body-frame accelerations and r the yaw rate from the IMU. The filter steps at the IMU period T.

2 Continuous model

$$\dot{p}_x = v_x, \qquad \dot{p}_y = v_y, \qquad \dot{\theta} = r, \qquad \dot{v}_x = \cos\theta\, a_x - \sin\theta\, a_y, \qquad \dot{v}_y = \sin\theta\, a_x + \cos\theta\, a_y. \tag{2}$$

3 Discrete map (Euler)

$$\boldsymbol{x}_{k+1} = \boldsymbol{f}_d(\boldsymbol{x}_k, \boldsymbol{u}_k) = \boldsymbol{x}_k + T \begin{bmatrix} v_x \\ v_y \\ r \\ \cos\theta\, a_x - \sin\theta\, a_y \\ \sin\theta\, a_x + \cos\theta\, a_y \end{bmatrix}, \qquad \theta \text{ wrapped to } [-\pi, \pi). \tag{3}$$

4 Jacobians of the discrete map

Evaluated at the estimate before propagation and at the current IMU sample:

$$\boldsymbol{F}_k = \frac{\partial \boldsymbol{f}_d}{\partial \boldsymbol{x}} = \begin{bmatrix} 1 & 0 & 0 & T & 0 \\ 0 & 1 & 0 & 0 & T \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -T(\sin\theta\, a_x + \cos\theta\, a_y) & 1 & 0 \\ 0 & 0 & T(\cos\theta\, a_x - \sin\theta\, a_y) & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{G}_k = \frac{\partial \boldsymbol{f}_d}{\partial \boldsymbol{u}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T \\ T\cos\theta & -T\sin\theta & 0 \\ T\sin\theta & T\cos\theta & 0 \end{bmatrix}. \tag{4}$$

5 Process noise

$$\boldsymbol{\Sigma}_u = q_{\text{scale}}\ \text{diag}(\sigma_a^2, \sigma_a^2, \sigma_r^2), \qquad \boldsymbol{Q}_k = \boldsymbol{G}_k \boldsymbol{\Sigma}_u \boldsymbol{G}_k^{\mathsf{T}} + \boldsymbol{Q}_{\text{floor}}, \tag{5}$$

$$\boldsymbol{Q}_{\text{floor}} = \text{diag}(\epsilon,\ \epsilon,\ q_\theta,\ q_v,\ q_v). \tag{6}$$

σ_a, σ_r are the per-sample IMU white-noise standard deviations, inflated by q_{scale}. q_v and q_θ are the per-step process-noise variances on velocity and heading: tuning parameters, chosen so that the filter is consistent (time-averaged NEES near 5 with the marker visible). They express how far the velocity and heading predictions may drift from the IMU integration between fixes, whether through residual bias, slip or target motion.

6 Predictor (every IMU sample)

$$\hat{\boldsymbol{x}}_k^- = \boldsymbol{f}_d(\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_k), \qquad \boldsymbol{P}_k^- = \boldsymbol{F}_k \boldsymbol{P}_{k-1} \boldsymbol{F}_k^{\mathsf{T}} + \boldsymbol{Q}_k. \tag{7}$$

7 Fiducial measurement (15 Hz, captured at $t_c = t_k - \tau_d$)

Pre-transform of the raw solver output, the planar translation $\boldsymbol{t}$ and yaw ψ of the marker in the camera frame, with $\boldsymbol{m}$ the marker position in T and $\boldsymbol{l}_c$ the camera position in the body frame:

$$\theta_z = \psi_0 - \psi, \qquad \boldsymbol{p}_z = \boldsymbol{m} - \boldsymbol{R}(\theta_z)\,(\boldsymbol{t} + \boldsymbol{l}_c). \tag{8}$$

The pseudo-measurement is then linear in the state:

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{p}_z \\ \theta_z \end{bmatrix} = \boldsymbol{H}_f \boldsymbol{x} + \boldsymbol{v}, \qquad \boldsymbol{H}_f = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \tag{9}$$

$$\boldsymbol{R}_f = \text{diag}(\sigma_p^2, \sigma_p^2, \sigma_\theta^2), \qquad \sigma_p = \sigma_{p,0}\sqrt{1 + (d/d_0)^2} \quad \text{with } d \text{ the range to the marker.} \tag{10}$$

8 Corrector (when the fix becomes available), with delay compensation

$$\boldsymbol{x}_{\text{ref}} = \text{buffer entry at the capture time } t_c \quad (\text{or } \hat{\boldsymbol{x}}_k^- \text{ if compensation is off}), \tag{11}$$

$$\boldsymbol{\nu} = \boldsymbol{z} - \boldsymbol{H}_f \boldsymbol{x}_{\text{ref}}, \qquad \nu_3 \text{ wrapped}, \tag{12}$$

$$\boldsymbol{S} = \boldsymbol{H}_f \boldsymbol{P}_k^- \boldsymbol{H}_f^{\mathsf{T}} + \boldsymbol{R}_f, \tag{13}$$

$$\boldsymbol{K} = \boldsymbol{P}_k^- \boldsymbol{H}_f^{\mathsf{T}} \boldsymbol{S}^{-1}, \tag{14}$$

$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}\boldsymbol{\nu}, \qquad \theta \text{ wrapped}, \tag{15}$$

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H}_f)\,\boldsymbol{P}_k^-\,(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H}_f)^{\mathsf{T}} + \boldsymbol{K}\boldsymbol{R}_f\boldsymbol{K}^{\mathsf{T}} \qquad \text{(Joseph form)}. \tag{16}$$

9 Fiducial outlier gate

Each fix is accepted only if $\boldsymbol{\nu}^{\mathsf{T}} \boldsymbol{S}^{-1} \boldsymbol{\nu} \le 100$ (about ten standard deviations); otherwise it is discarded and the state and covariance are left unchanged. The threshold is deliberately loose: it rejects a wrong-marker or degenerate pose solve, metres off, but never a plausible fix, so an unmodelled change such as a target starting to move is still absorbed rather than rejected.

RTK position and INS heading are not used by the docking filter. In the navigation filter, which works in the world frame, they are the rows $\boldsymbol{z}_W = [\boldsymbol{p}_{\text{gps}};\ \psi_W]$, $\boldsymbol{H} = [\boldsymbol{I}_3 \ \ \boldsymbol{0}_{3\times 2}]$, $\boldsymbol{R} = \text{diag}(\sigma_{\text{rtk}}^2, \sigma_{\text{rtk}}^2, \sigma_\psi^2)$, heading innovation wrapped.

10 Initialisation

First fiducial fix, velocity unknown (deployment form, both variants):

$$\hat{\boldsymbol{x}}_0 = \begin{bmatrix} \boldsymbol{z}_f \\ \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{P}_0 = \mathrm{blkdiag}(\boldsymbol{R}_f,\ \sigma_{v,0}^2 \boldsymbol{I}_2). \tag{17}$$

Handover from the navigation estimate (full-scale alternative, needs the target's reported pose):

$$\hat{\boldsymbol{x}}_0 = \begin{bmatrix} \boldsymbol{R}(\theta_t)^\mathsf{T}(\hat{\boldsymbol{p}}_W - \boldsymbol{p}_{t,W}) \\ \hat{\theta}_W - \theta_t \\ \boldsymbol{R}(\theta_t)^\mathsf{T} \hat{\boldsymbol{v}}_W \end{bmatrix}, \qquad \boldsymbol{T}_m = \mathrm{blkdiag}\big(\boldsymbol{R}(\theta_t)^\mathsf{T},\ 1,\ \boldsymbol{R}(\theta_t)^\mathsf{T}\big), \tag{18}$$

$$\boldsymbol{P}_0 = \boldsymbol{T}_m \boldsymbol{P}_W \boldsymbol{T}_m^\mathsf{T} + \mathrm{blkdiag}(\boldsymbol{P}_{\mathrm{tgt}}, \boldsymbol{0}). \tag{19}$$

11 Schedule, per IMU sample k

1. Predict (Section 6), unless $k = 1$.
2. For every fiducial fix with $t_{\mathrm{avail}} \le t_k$ not yet used: if valid, gate (Section 9) and correct (Section 8).
3. Store $\hat{\boldsymbol{x}}_k$ in the ring buffer at index k, *after* the corrections, so that a later delayed fix sees the best estimate of time t_k.
4. Log $\hat{\boldsymbol{x}}_k$ and $\mathrm{diag}(\boldsymbol{P}_k)$. With truth available, $\mathrm{NEES}_k = \boldsymbol{e}_k^\mathsf{T} \boldsymbol{P}_k^{-1} \boldsymbol{e}_k$ with $\boldsymbol{e}_k = \hat{\boldsymbol{x}}_k - \boldsymbol{x}_k$, heading component wrapped.

Two reminders

- The mean always goes through $\boldsymbol{f}_d$, never through $\boldsymbol{F}\hat{\boldsymbol{x}} + \boldsymbol{G}\boldsymbol{u}$.
- $\boldsymbol{F}_k$ and $\boldsymbol{G}_k$ are the derivatives of whatever $\boldsymbol{f}_d$ is actually implemented. With the second-order map $\boldsymbol{p}_{k+1} = \boldsymbol{p}_k + T\boldsymbol{v}_k + \frac{T^2}{2}\boldsymbol{R}(\theta_k)\boldsymbol{a}_k$ instead of Euler, the position rows of $\boldsymbol{F}_k$ gain $\frac{T^2}{2}\,\partial(\boldsymbol{R}\boldsymbol{a})/\partial\theta$ in the heading column and the position rows of $\boldsymbol{G}_k$ gain $\frac{T^2}{2}\boldsymbol{R}(\theta_k)$ in the accelerometer columns.